DePIN Verification Index – AGV Protocol v2025.10

(Field Documentation, Power Generation Proofs & Oracle Anchoring)

I. Field Documentation – Photos, Videos & Hash Proofs

No.	File / Folder	Description / Proof Scope
A.6-0	**Land Overview**	Site overview and boundary mapping for the 6.42 MW Agri-PV base in Yichuan.
A.6-1	**Land Leveling & Fertilization**	Ground preparation record before PV rack installation.
A.6-2	**Trellis Infrastructure**	Photographs of PV mounting racks and orchard trellis integration.
A.6-3	**Drip Irrigation System**	On-site irrigation and fertigation control room.
A.6-4	**Dwarf Dense Plantation**	Proof of agricultural layer and sustainable planting density.
A.6-5	**Plant Protection & Maintenance**	Crop inspection and orchard maintenance images.
A.6-6	**Agri-PV Integration**	Drone photos showing parallel deployment of orchard and thin-film PV arrays.
A.6-H	**Hash Anchoring for Field Documentation**	IPFS / Arweave hash records ensuring tamper-proof image verification.

II. Power Generation Data & Oracle Proofs

1

Daily Power Generation Snapshot (2025-10-02)

96-point IoT data sample; grid-connected 3.87 MWp station SX-YA; CSV + SHA-256 hash record.

2

Monthly Settlement & Reconciliation Certificate (2025-10)

Reconciles daily CSV totals with State Grid billing; signed by Ops, Finance, Legal, and Oracle Admin.

3

Oracle Hash Logs

Aggregated monthly SHA-256 records for CSV / Settlement / Bank Slip / Supporting Evidence.

4

Grid Settlement Invoice (Attached)

State Grid billing proof; cross-checked with on-chain hash for verification.

III. IoT / Oracle Architecture Summary

Layer	Function	Status
Sensor → Edge Gateway	Collects power-meter & inverter data every 15 min (96 points/day)	✅ Live
Gateway → Oracle	Aggregates CSV, signs SHA-256, pushes to blockchain (Power-to-Mint)	✅ Operational
Oracle → Chain	Writes verified hash to AGV Protocol smart contract	✅ Connected
Chain → Dashboard	DAO Dashboard displays verified energy yield for token mint	🔄 Dashboard Alpha

IV. External Verification & Certification

Field Audit

AGV Internal Audit Team

Verified location, coordinates, and operation logs.

IoT Oracle Validation

DAO Operations / Technical Lead

Confirmed data signing & on-chain hash entries.

Translation & Legal Proof

NAATI Certified Translator #3327-620583-6

Ensures English–Chinese translations of energy data carry legal validity.